

STAT lecture 4

Simultaneous random variables, covariance and correlation

Disclaimer:

**Please note that these slides are only a brief overview of the topic.
Many more details will be explained on the blackboard and with
MATLAB examples**

Simultaneous random variables

Stochastic experiment

Random variables, X and Y , measures two quantities for each outcome in sample space, that is $(X, Y): S \rightarrow \mathbb{R}^2$

Generalizes straightforward to more than two variables

Discrete (X, Y)

Simultaneous pmf $f_{X,Y}(x, y) = P(X = x \wedge Y = y)$

Marginal pmf's $f_X(x) = \sum_y f_{X,Y}(x, y)$

$$f_Y(y) = \sum_x f_{X,Y}(x, y)$$

Conditional pmf's $f_{X|Y}(x) = \frac{f_{X,Y}(x, y)}{f_Y(y)}$

$$f_{Y|X}(y) = \frac{f_{X,Y}(x, y)}{f_X(x)}$$

Continuous (X,Y)

Simultaneous pdf

$$f_{X,Y}(x, y) \geq 0$$

$$\iint_{\mathbb{R}^2} f_{X,Y}(x, y) dx dy = 1$$

$$P((X, Y) \in A \subseteq \mathbb{R}^2) = \iint_A f_{X,Y}(x, y) dx dy$$

Marginal pdf's

$$f_X(x) = \int_{\mathbb{R}} f_{X,Y}(x, y) dy$$

$$f_Y(y) = \int_{\mathbb{R}} f_{X,Y}(x, y) dx$$

Conditional pdf's

$$f_{X|Y}(x) = \frac{f_{X,Y}(x, y)}{f_Y(y)}$$

$$f_{Y|X}(y) = \frac{f_{X,Y}(x, y)}{f_X(x)}$$

Independent X, Y

$$f_{X,Y}(x, y) = f_X(x) f_Y(y)$$

Covariance

How are X and Y varying together ?

Definition $Cov(X, Y) = \sigma_{XY} \stackrel{\text{def}}{=} E[(X - \mu_X)(Y - \mu_Y)] = E[XY] - \mu_X\mu_Y$

Interpretation

$Cov(X, Y) > 0$	When X increases, Y also increases <i>in the mean sense</i>
$Cov(X, Y) < 0$	When X increases, Y decreases <i>in the mean sense</i>
$Cov(X, Y) \cong 0$	X and Y are uncorrelated

The covariance is unnormed = *not* invariant to change of units/scale

Correlation

Normed covariance = invariant to change of units/scale

Definition $\rho_{XY} \stackrel{\text{def}}{=} \frac{Cov(X, Y)}{\sqrt{V(X)}\sqrt{V(Y)}} = \frac{\sigma_{XY}}{\sigma_X\sigma_Y}, \quad -1 \leq \rho_{XY} \leq 1$

Linear combinations of random variables

$$Y \stackrel{\text{def}}{=} \sum_{i=1}^n c_i X_i \quad \Rightarrow \quad E[Y] = \sum_{i=1}^n c_i E[X_i] \quad \text{and}$$

$$V[Y] = \sum_{i=1}^n c_i^2 V[X_i] + \sum_{i=1}^n \sum_{j \neq i} c_i c_j \text{Cov}[X_i, X_j]$$

Special case $X_1, \dots, X_n$ independent: $V[Y] = \sum_{i=1}^n c_i^2 V[X_i]$

Often used linear combination (average of observations)

$$\bar{X} \stackrel{\text{def}}{=} \frac{1}{n} \sum_{i=1}^n X_i \quad \text{with independent } X_i, \quad E[X_i] = \mu, \quad V[X_i] = \sigma^2$$

$$E[\bar{X}] = \mu$$

$$V[\bar{X}] = \frac{\sigma^2}{n}$$

General (non-linear) function of random variable

- Given
- random variable X with pdf $f_X(x)$
 - non-linear *monotonous (invertible)* function $y = h(x)$
 - new random variable $Y = h(X)$

Random variable Y has pdf

$$f_Y(y) = \left| \frac{dh^{-1}(y)}{dy} \right| f_X(h^{-1}(y))$$